



6. Overfitting and Underfitting

When being too good is a bad sign

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Overfitting and Underfitting are two critical pitfalls when studying Machine Learning that tells you:

| ***“If your model are studying well... or just pretending to.”***

Let me explain it the way I usually see it.

1. Overfitting — The Illusion of Perfection

At first glance, overfitting looks impressive. A model achieves **99%–100% accuracy** on training data and you might think: *“Wow... this model is brilliant”*.

But here’s the uncomfortable truth: **Perfection in training is often a red flag.**

Overfitting happens when a model learns the training data **too well**—not just the real patterns, but also the **noise, randomness, and tiny details** that don’t

actually matter.

It's like a student who memorizes an entire textbook word for word. They can answer every question from that book flawlessly. But the moment you give them something slightly different? They freeze. Because they didn't *understand*. They just *remembered*.

A model suffering from overfitting behaves exactly the same way:

- It performs **extremely well on training data (high accuracy on training data)**
- But performs **poorly on validation or testing data (low accuracy on testing data)**
- There is a **large gap** between training and testing performance
- It captures even **random fluctuations** instead of true patterns

Why does Overfitting happen? There are several reasons, but they all point to one idea: **The model becomes too sensitive.**

- **High Variance, Low Bias** → It reacts too much to small changes
- **Model too complex** → Too many parameters, too much flexibility
- **Too many features** → Not all of them are meaningful
- **No proper validation split** → The model never gets tested on new data
- **Training too long** → It starts learning noise after learning patterns
- **Too little data** → Not enough examples to generalize properly

In short: **The model memorizes instead of learning.**

2. Underfitting — The Failure to Learn

Now let's go to the opposite extreme. Underfitting is not impressive at all. In fact, it's painfully obvious. The model performs poorly on **both training and testing data**.

This means one thing: **It didn't even learn the basics.**

Think of a student who doesn't study properly. They don't remember the vocabulary... They don't understand the meaning. And when faced with new words? They fail again. This is underfitting. The model is too simple, too limited, or too poorly trained to capture the underlying structure of the data.

Signs of underfitting:

- **Low accuracy on training data**
- **Low accuracy on testing data**
- Errors are **high and similar** across both
- The model fails to capture **any meaningful pattern**

Why does Underfitting happen? This time, the problem is the opposite: **The model is too weak.**

- **Low Variance, High Bias** → Too rigid to adapt
- **Model too simple** → Not enough capacity
- **Poor feature selection** → Inputs don't help prediction
- **Insufficient data** → Not enough to learn from
- **Unscaled features** → Learning becomes difficult
- **Training stopped too early** → Didn't learn enough
- **Wrong algorithm** → Example: linear model for non-linear data

In short: **The model fails to understand the data at all.**

3. The Balance — What a “Good” Model Looks Like

So if overfitting is “too good”, and underfitting is “not good enough”. What are we actually aiming for? A good Machine Learning model lives **in between**.

It should:

- Learn meaningful patterns from training data
- Perform **consistently well** on unseen data
- Avoid both memorization and ignorance

In other words: **It understands, and it generalizes.**

Let me put it in the simplest way possible:

- **Overfitting** → A student who memorizes everything but cannot think
- **Underfitting** → A student who understands nothing and cannot learn
- **Good Model** → A student who understands concepts and adapts to new problems

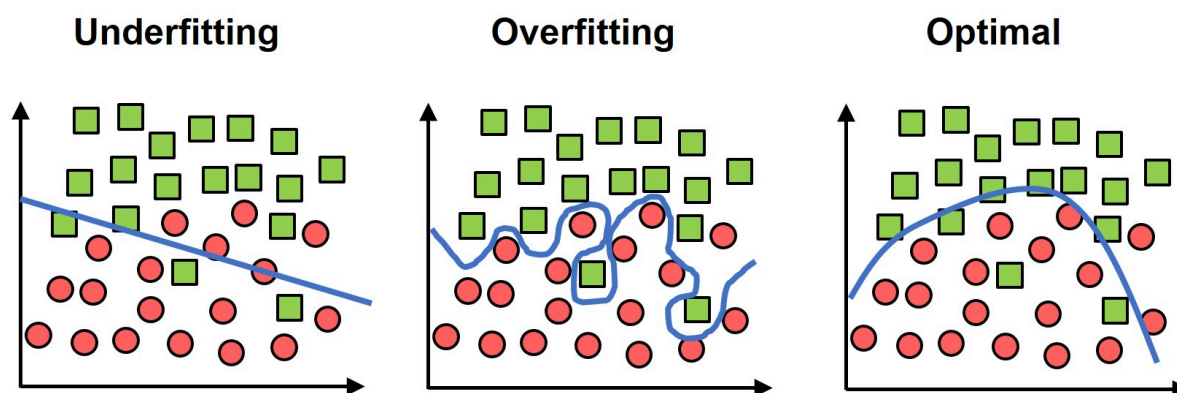
And that's the real goal of Machine Learning: Not perfection. Not memorization. But **understanding that survives the unknown**.

4. A Geometric Way to See Overfitting, Underfitting, and Right Fitting

Let's pause for a moment and *visualize* everything we just discussed. Imagine you are given a set of data points belonging to two different classes.

Your task is simple: **Draw a boundary that separates them**. But *how* you draw that boundary... reveals everything.

Here is how the boundary is drawn for each of the three cases:



Let's break it down:

The Balanced Model — “Understanding the Pattern”: A good model doesn't try to be perfect. It tries to be **correct in general**. So instead of forcing every single point to be classified perfectly, it draws a boundary that **captures the overall structure of the data**.

- The line (or curve) separates the two classes **clearly**
- A few points might fall on the wrong side → and that's okay
- Because those points might be **noise or exceptions**

This is what we call **generalization**. The model understands the *pattern*, not the *exceptions*.

Overfitting — “Trying Too Hard”: Now imagine you try to classify **every single point correctly**. So what do you do?

You start bending the boundary, twisting it, making it more and more complex. Until it perfectly wraps around every point.

- The boundary becomes **highly irregular**
- It looks like a **wiggly, aggressive curve**
- Every training point is classified correctly

And yet, this is exactly the problem. Because this boundary is not learning the true pattern. It is just **memorizing the data points**. The model is no longer separating classes, it is *tracing them*.

Underfitting — “Not Trying Enough”: Now go to the other extreme. You decide: *“I’ll just draw a simple straight line”*.

No matter how the data is distributed.

- The boundary is too **simple**
- It fails to separate the classes properly
- Many points are misclassified

This model doesn’t even try to capture the real structure → It ignores the pattern completely.

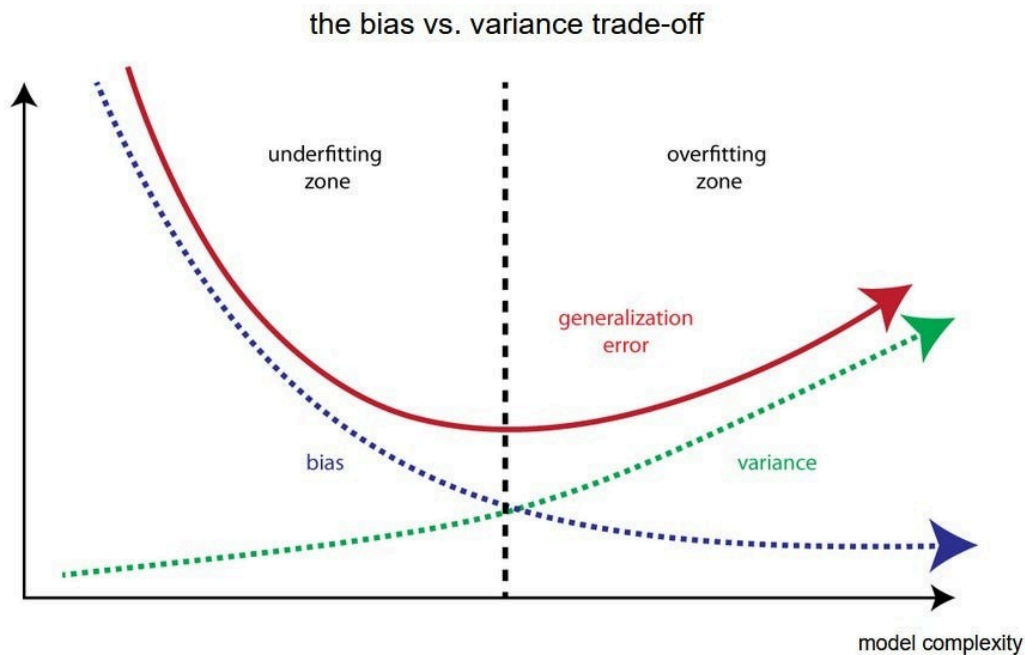
If I had to summarize this in one sentence:

- **Overfitting** → “Let me pass through every point.”
- **Underfitting** → “Let me ignore most of the points.”
- **Balanced Model** → “Let me understand the shape of the data.”

And this picture, this simple act of drawing a boundary, is exactly what leads us to the next question: **Why does a model behave this way?** And that’s where **Bias and Variance** finally come in.

5. The Bias-Variance Trade-Off

Variance and Bias - The hidden indicators behind Overfitting and Underfitting. If overfitting and underfitting are the *symptoms*, then **Bias and Variance are the real diagnosis**. They tell us *why* a model behaves the way it does.



Two Ways a Model Can Be Wrong

Every Machine Learning model makes errors. But those errors usually come from **two very different sources**:

- **Bias — “How wrong am I on average?”**: Bias measures how far your model's predictions are from the true values.
 - **High Bias** → The model is too simple
 - It makes strong assumptions
 - It consistently misses the real patterns
 → This leads to **underfitting**.

Think of it like this: A student who oversimplifies everything. “Every essay has the same structure... every problem has the same solution”. They are not flexible enough to understand the problem.

- **Variance — “How unstable am I?”**: Variance measures how much your model changes when trained on different data.
 - **High Variance** → The model is too sensitive
 - Small changes in data → big changes in predictions
 - It captures noise instead of general patterns

→ This leads to **overfitting**.

This is like a student who overreacts to every detail. *"If this one example looks like this... then everything must follow it"*. They don't generalize—they over-specialize.

Reading Bias & Variance Through Performance - We usually don't calculate bias and variance directly. Instead, we **observe them through training vs testing performance**:

Scenario	Bias	Variance	Interpretation
Low training error + Low testing error	Low	Low	Good model
High training error + High testing error	High	Low	Underfitting
Low training error + High testing error	Low	High	Overfitting

Examples (The "Reality Check"). Let's make this concrete:

- **99% (train) → 27% (test)**
→ Low Bias, High Variance → **Overfitting**
- **30% (train) → 17% (test)**
→ High Bias, Low Variance → **Underfitting**
- **95% (train) → 92% (test)**
→ Low Bias, Low Variance → **Balanced Model**

A Simple Rule You Should Never Forget: If a model performs poorly on training data, it will perform even worse on testing data.

Why? Because training data is what it already *knows*. Testing data is what it has *never seen*. So if it struggles with the familiar, it has no chance with the unknown.

The Tolerance Gap — How Much Difference is Acceptable? Because not every gap between training and testing is bad. We expect *some difference*. But how much is too much?

- **3–5% difference** → Acceptable (well-balanced)
- **10–15% difference** → Likely overfitting
- **Both below ~60–70%** → Likely underfitting (regardless of gap)

This "gap" is one of the easiest ways to **diagnose your model quickly**.

Fixing the Problem — Based on the Cause

Now comes the important part: **You don't fix symptoms, you fix causes.**

- **Fixing Overfitting (High Variance)** - Goal: **Make the model less sensitive**

- Use a **simpler model**
- Reduce number of features (feature selection)
- Apply **regularization** (Lasso, Ridge)
- Collect **more data**
- Use **cross-validation**
- Apply **early stopping**
- Use **data augmentation**

→ Idea: *Stop the model from memorizing*

- **Fixing Underfitting (High Bias)** - Goal: **Make the model more expressive**

- Use a **more complex model**
- Add more **relevant features**
- Reduce excessive **regularization**
- Improve **data quality**
- Train for **longer**
- Apply **feature scaling**
- Add **polynomial / interaction features**

→ Idea: *Help the model actually learn*

Let me simplify everything into one picture:

- **High Bias (Underfitting)** → The model is *blind*
- **High Variance (Overfitting)** → The model is *overreacting*
- **Balanced Model** → The model *understands and adapts*

Bias and Variance are not just technical terms.

They describe a deeper truth:

- Learn too little → You miss everything
- Learn too much → You trust everything
- Learn just enough → You understand what matters

And that is where Machine Learning actually works.

Summary

Overfitting and Underfitting is like **how a student studies for an exam**.

- **Overfitting** is the student who memorizes everything word-for-word. They get perfect scores on practice questions, but fail the moment the questions change.
- **Underfitting** is the student who doesn't study enough. They don't understand the material, and perform poorly on both familiar and new questions.
- **A good model** is the student who understands the concepts. They may not be perfect, but they can adapt and solve new problems.

In the end, Machine Learning is not about being perfect on what you've seen. It's about being **reliable on what you haven't**.

References

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 [7. Supervised & Unsupervised Learning](#)